Bioinformatic analysis of het-siRNAs and phasiRNAs in the symbiotic interaction between *Phaseolus vulgaris* and *Rhizobium etli*

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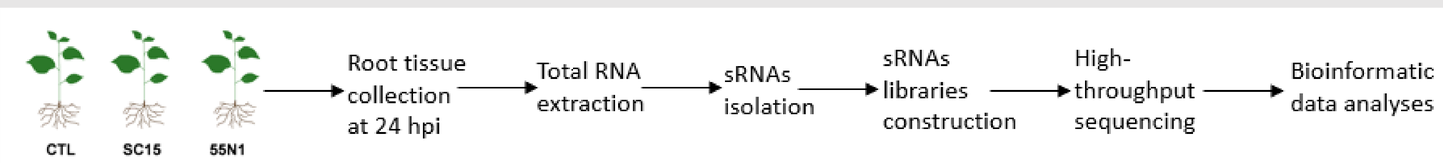
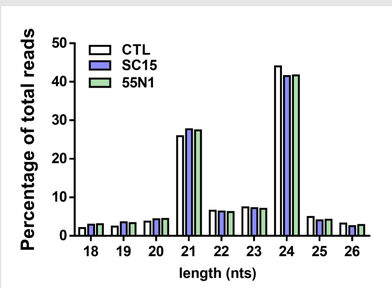
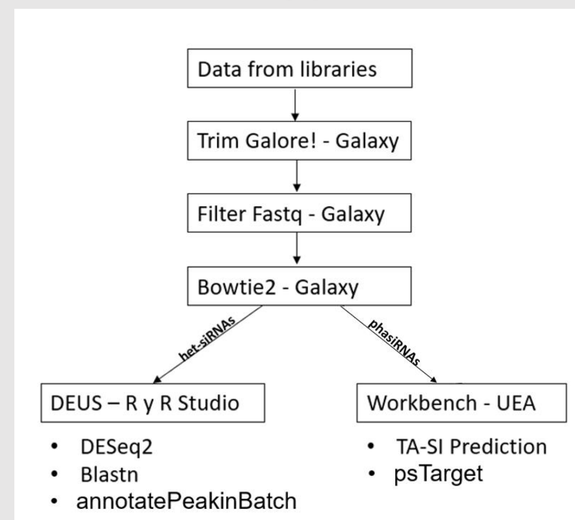
Small RNAs (sRNAs) act as regulators of gene expression at transcriptional and post-transcripcional levels. One of the largest and best characterized group of sRNAs are small interference RNAs (siRNAs), which are processed from double stranded RNA precursors to produce mature RNAs of 20-24 nucleotides that play different regulatory functions. Heterochromatic siRNAs (het-siRNAs) are 23-24 nt siRNAs that mediate *de novo* DNA methylation and chromatin accessibility. On the other hand, phased secondary small interfering RNAs (phasiRNAs) are 21 nt in length and their production is triggered by the endonucleolytic cleavage of a miRNAs, leading to the post-transcriptional regulation of target transcripts in *cis* or *trans*. In this work, we characterized het-siRNAs and phasiRNAs differentially regulated in the nitrogen-fixing symbiosis between common bean (*Phaseolus vulgaris*) and two strains of its symbiont *Rhizobium etli* that exhibit different symbiotic efficiency.

Library preparation and bioinformatic strategy





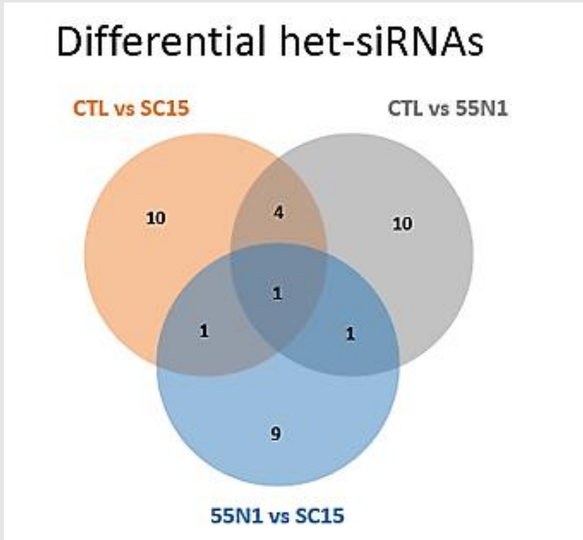
Plants were inoculated with the more efficient *R. etli* strain SC15, the less efficient strain 55N1 or with the media used to growth *R. etli* (CTL). sRNAs were isolated and used to prepare and sequence Illumina libraries. Total reads were classified according to their size for each condition.

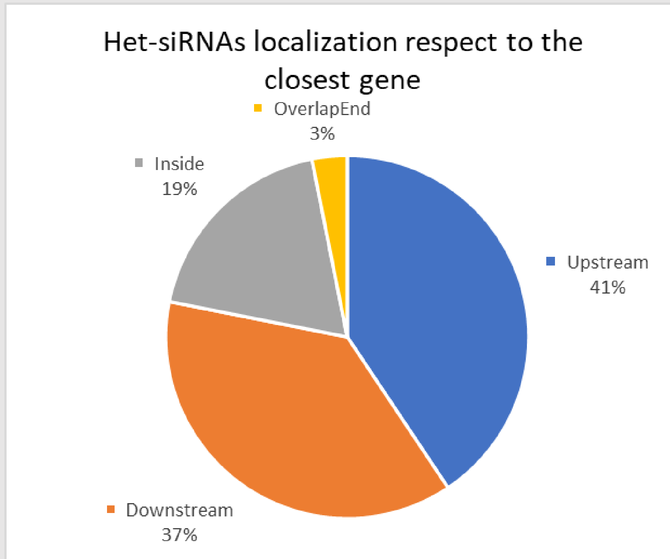


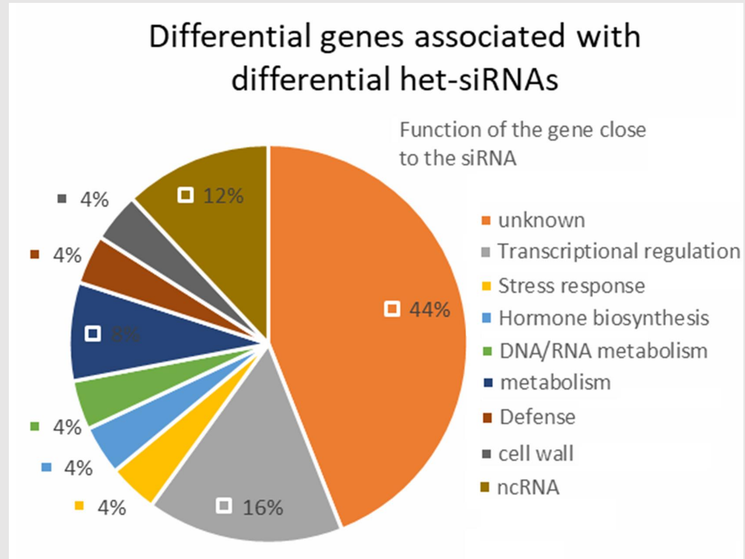
Adapters were removed with the Trim Galore Tool. Sequences were filtered with Filter FastQ using 23-24 nucleotides for het-siRNAs and 21-22 nucleotides for phasiRNAs. The rRNA and tRNA were removed using Bowtie2. Het-siRNAs were quantified using DEUS. PhasiRNAs were identified using TA-SI prediction tool of Workbench. Putative targets in the *P. vulgaris* genome were identified with psTarget.

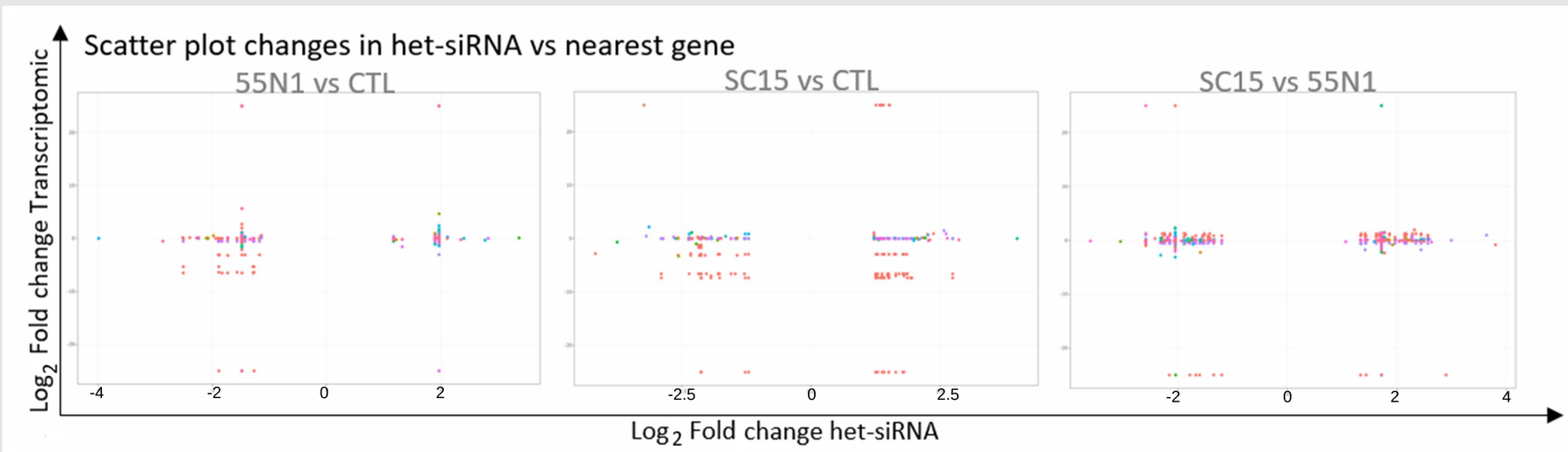
het-siRNAs

We identified 236 het-siRNAs differentially accumulated in all pairwise comparisons. Twenty-five of these het-siRNA peaks are located near differentially regulated genes, which encode proteins with unknown functions (44%), transcription factors (16%) or non-coding RNAs (12%).





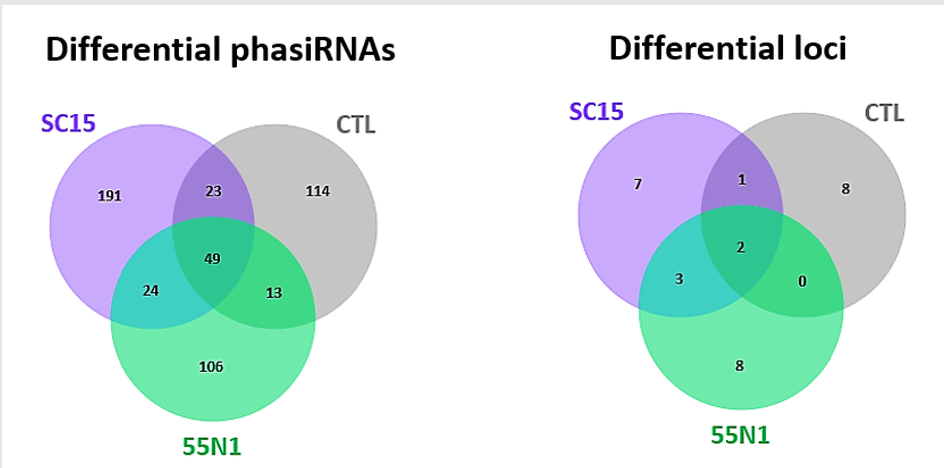


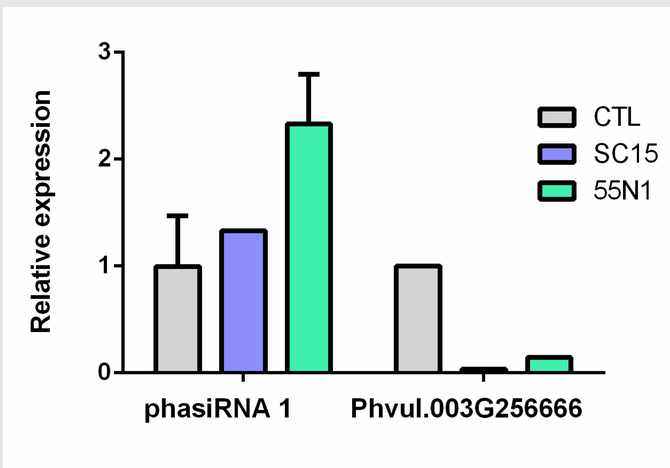


Scatter plot showing changes in abundance of het-siRNAs and transcripts of their nearest genes according to *P. vulgaris* transcriptomic data generated in our lab.

phasiRNAs

678 phasiRNAs corresponding to 18 loci were differentially regulated. Putative targets of differential phasiRNAs (diiferential loci) include receptors, transcription factors and cytoskeleton components.





Phasi #1 targets *Phvul.003G256666*, which encodes an RNA polymerase. Whereas Phasi #1 is upregulated by infection with rhizobia, its putative target is downregulated.

This work was supported by PICT 2016-0333 and 2017-00069